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Honor Pledge

I, **Poornima Madhu M**, **CB.EN.U4CCE23036**, do hereby declare that I have neither taken nor provided any help, from any source, for the purpose of completing this assignment. I understand that typing my full name in the space below, is equivalent to affixing my signature to this statement and that it is binding.

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Subject-Independent Ictal versus Interictal Classification from Single-Lead Wearable ECG on SeizeIT2

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***Abstract*—ECG studies often mix one patient's segments into both training and test data, which can inflate scores. We compared ictal and interictal single-lead ECG segments from 119 SeizeIT2 patients under subject-independent and within-patient splits. Four classical classifiers used HRV, Poincaré and morphology features; two 1D CNNs used raw windows. On 24 unseen patients a random forest reached F1 0.895, against 0.373 for the best CNN. A within-patient split raised it to 0.941. Ictal segments were shorter than interictal ones, so this is state separability, not seizure prediction.**

***Index Terms*—electrocardiography, epilepsy, heart rate variability, random forest, SeizeIT2, subject-independent evaluation**

I. Introduction

Seizures are hard to anticipate, and scalp EEG is difficult to keep on for long periods outside a hospital. A seizure also disturbs the autonomic nervous system, so heart rate, heart-rate variability (HRV) and ECG morphology change, and a single-lead ECG sensor is easier to wear than an EEG cap [1], [2]. The open SeizeIT2 dataset, released by KU Leuven, offers such wearable recordings from 125 patients with focal epilepsy: 11,640 hours in total, with 886 annotated seizures.

Clinicians describe a seizure in four phases: interictal (between seizures), pre-ictal (before onset), ictal (the seizure itself) and post-ictal (recovery). Detecting a seizure that is under way is a different problem from predicting one that has not started. Prediction needs the pre-ictal phase, whereas detection, and the classification task in this paper, only needs to tell the ictal state from a quiet one. Cardiac changes can be measured in all of these phases, which is why ECG is studied for both problems. The question we ask is narrower: how well can ictal and interictal ECG segments be told apart, and how much of that depends on the way the data are split?

Many ECG studies report results from one random split. A recent analysis of ECG-based prediction notes that variability between patients and within a patient makes such results hard to compare [3]. If segments from one patient land in both the training and the test set, a classifier can learn that patient's cardiac signature instead of anything seizure-related. We could not find a direct measurement of this effect for ECG on SeizeIT2.

We make three contributions.

- A pipeline that turns SeizeIT2 EDF recordings into labelled ictal and interictal ECG segments and extracts HRV, Lorenz/Poincaré and morphology features (Section III).
- A side-by-side comparison of a subject-independent and a within-patient split on the same feature table, with four classical classifiers (Section IV).
- A comparison with a one-dimensional CNN trained on raw ECG windows. Under the same subject-independent split it scores far below the feature-based random forest.

The task is to classify segments that are already known to be ictal or interictal. It is not seizure prediction, and none of our numbers is a warning time or a false-alarm rate. Section V explains a duration confound that limits what the scores can show.

II. Literature Survey

Classical machine learning on ECG features. Seifi *et al.* trained SVM, k-NN and naive Bayes classifiers on 13 ECG features from 13 people in the Siena data and found naive Bayes the most suitable [1]. Abtahi *et al.* used heart-rate-variability (HRV), Lorenz-plot and multifractal ECG features with a random forest and SHAP on the EPILEPSIAE data, and reported 85.9 % accuracy and 84.6 % sensitivity [2]. Our feature families follow their work. Vishar *et al.* also applied a random forest, to pre-ictal stage prediction [4].

Signal-processing and deep-feature approaches. Perez-Sanchez *et al.* combined a maximal-overlap discrete wavelet packet transform and a homogeneity index with a k-NN classifier on MIT-BIH data, reporting 93.25 % accuracy for a 20-min prediction interval [5]. Muralidharan *et al.* used spectral whitening to remove normal cardiac, respiratory and brain activity from the ECG before classification [6]. Abarna *et al.* used wavelet and variational mode decomposition [7]. Muralidharan *et al.* trained a CNN with a joint cross-entropy and center loss on the TUSZ corpus, and reported an AUROC of 0.8327 [8]. These figures are as given in the cited papers and use different data and windows, so they cannot be compared with each other or with our results.

EEG–ECG and deep models. Phomsiricharoenphant *et al.* studied EEG and ECG together with the Hilbert–Huang transform [9]. Subash *et al.* proposed a framework for early seizure detection [10], and Phuyal and Morshed used an adversarial vision transformer with Rényi entropy on wearable EEG and ECG [11]. Truong *et al.* showed that convolutional networks can learn seizure-related patterns from intracranial and scalp EEG [12]. We test whether a comparable raw-signal network works on single-lead ECG.

Evaluation. Kalousios *et al.* trained patient-specific models on 97 HRV features from 39 patients with 252 seizures. They reported a mean ROC-AUC of 0.823 with non-causal evaluation and 0.569 with pseudo-prospective evaluation, and pointed to variability between and within patients [3]. This gap between evaluation schemes is the reason we compare split protocols.

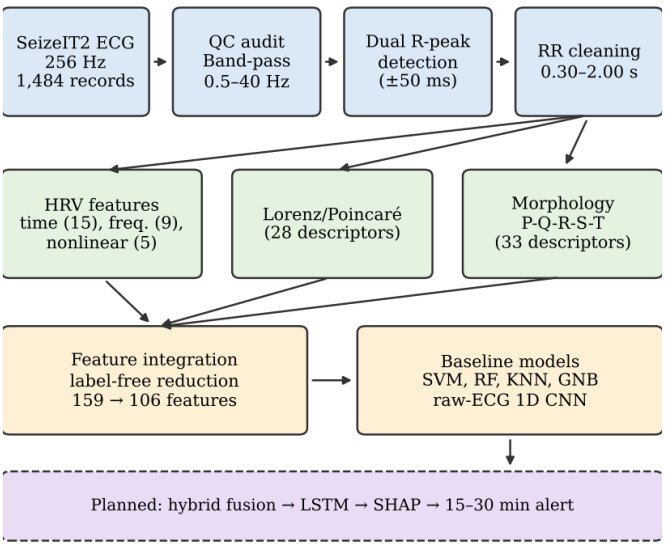


Fig. 1. ECG processing pipeline. Solid boxes indicate implemented stages, whereas the dashed box indicates the planned hybrid stage.

TABLE I
SEGMENT DATASET USED IN THIS STUDY.

Class	Segments	Patients	Duration (s)
Ictal	851	119	mean 58.6, median 33, range 2–962
Interictal	822	119	300 (fixed)
Total	1,673	119	

Research gap. The studies above differ in data, window length and validation. In the papers we reviewed we found none that measures how much a patient-mixed split inflates ECG results on SeizeIT2. We measure that gap.

III. METHODS

Fig. 1 shows the pipeline. Everything was implemented in Python.

A. Data and Segment Extraction

SeizeIT2 contains wearable recordings from 125 patients with focal epilepsy, 11,640 h in total and 886 annotated seizures. The ECG is a single lead sampled at 256 Hz. Ictal segments were cut from the seizure onset and duration given in each recording's events file (Fig. 2). Interictal segments were fixed 5-min windows (300 s) taken at least 30 min from every annotated seizure in the same recording. A run that has an events file but no seizure annotation counts as seizure-free. This is an operational definition, not a guarantee.

We ended up with 851 ictal and 822 interictal segments from 119 patients, and every patient contributes both classes (Table I). The two classes differ sharply in length. Ictal segments have a mean duration of 58.6 s and a median of 33 s, with a range of 2–962 s; 63.8 % of them are shorter than 60 s and only 14 reach 300 s. This difference matters for the features and for the interpretation of the results, and we come back to it in Section V.

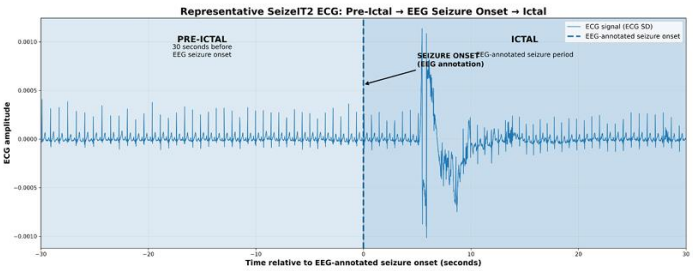


Fig. 2. Representative single-lead SeizeIT2 ECG 30 s before and after an EEG-annotated seizure onset (dashed line). The shaded part after onset is the kind of signal an ictal segment contains; interictal segments in this study lie at least 30 min away from any seizure.

B. Preprocessing and Feature Extraction

Each segment was filtered with a zero-phase fourth-order Butterworth band-pass (0.5–40 Hz). R-peaks were detected with the NeuroKit2 package, and RR intervals outside 0.30–2.00 s were discarded. Segments with fewer than 10 clean RR intervals were flagged, and frequency-domain HRV was computed only when at least 60 clean intervals spanning more than 60 s were available. Because many ictal segments are short, 411 segments were too short for frequency-domain HRV and 106 had too few RR intervals. We left those feature values missing instead of setting them to zero.

The feature table follows the usual HRV definitions and the feature families used for seizure prediction in [2]. It has five families: time-domain HRV (e.g., MeanNN, SDNN, RMSSD, pNN50), frequency-domain HRV (VLF, LF and HF power, LF/HF), nonlinear HRV (Poincaré SD1 and SD2, sample entropy, DFA), Lorenz/Poincaré return-map descriptors, and beat-level morphology descriptors (QRS width, amplitudes, QT approximation). We removed segment length and count-type quality counters from the classifier inputs. That left 95 candidate features.

Of the 95 candidates, 82 had at least 80 % non-missing values in both classes of the training data, and the filter was computed on training data only. The filter had a side effect that we want to state clearly. It removed all 8 frequency-domain and all 5 nonlinear HRV features, because they were missing for most ictal segments. Their training coverage was 0.97 in the interictal class but only 0.35 for the frequency-domain features and 0.32–0.56 for the nonlinear ones in the ictal class. The classifiers therefore see the 11 time-domain HRV, 27 Lorenz/Poincaré and 44 morphology features only.

C. Classical Classifiers

We built four scikit-learn pipelines. All use median imputation, and all except the random forest standardise the features: an RBF-kernel SVM ($C = 1$, balanced class weights), a random forest (500 trees, minimum leaf size 2, balanced subsample weights), k-nearest neighbours ($k = 5$, distance weighting) and Gaussian naive Bayes. We did not tune hyperparameters. The settings are the ones in our scripts, so the scores should be read as a first comparison and not as the best each model can do.

D. Evaluation Metrics

The ictal class is positive. With true and false positives and negatives written TP, FP, TN and FN, we report precision, recall and F_1 (1) and the Matthews correlation coefficient (MCC) (2):

$$\text{Prec.} = \frac{TP}{TP+FP}, \quad \text{Rec.} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2\,\text{Prec.}\,\text{Rec.}}{\text{Prec.} + \text{Rec.}}, \tag{1}$$

$$\text{MCC} = \frac{TP\,TN - FP\,FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}. \tag{2}$$

Balanced accuracy is the mean of recall and the recall of the interictal class. ROC-AUC is the area under the curve of recall against the false positive rate, and PR-AUC is the average precision. Accuracy alone would be misleading for the CNN windows, where only 16.8 % are ictal, so we also give recall, F1 and balanced accuracy.

E. Raw-Signal CNN

A one-dimensional CNN was trained on the extracted ECG without hand-crafted features, using TensorFlow. Segments were cut into 10 s windows with a 5 s stride at 256 Hz. Each window was centred on its median, scaled by its interquartile range and clipped to ± 10. The network has three convolution layers (32, 64 and 128 filters with kernel sizes 7, 5 and 3), batch normalisation and max pooling after the first two, global average pooling, dropout of 0.35, a 64-unit dense layer, dropout of 0.25 and a sigmoid output. We trained it with the Adam optimiser (learning rate 10^{-3}) and batch size 64. The baseline version used a 0.5 decision threshold. The improved version added class weights, a validation set, learning-rate reduction on plateau and a decision threshold chosen on the validation set to maximise ictal F1.

F. Experimental Setup

Protocols. In the *subject-independent* protocol, 20 % of subjects were held out for testing (scikit-learn `GroupShuffleSplit`, seed 42, first split containing both classes), giving 95 training and 24 test subjects, and 1,491 training and 182 test segments. The test set holds 91 segments per class. No subject appears on both sides. All three subject-wise experiments (classical models, baseline CNN, improved CNN) use the same 24 test subjects, so their scores are comparable. For the CNN, the test set expands to 6,315 windows, of which 1,064 are ictal and 5,251 interictal. The improved CNN further splits the 95 non-test subjects into 83 training and 12 validation subjects.

In the *within-patient* protocol, segments of each patient and class were split roughly 80:20 at random; a patient-class group with a single segment stayed in training. This gives 1,302 training and 371 test segments, with 93 patients present in both sets. We report it only to show how optimistic such a split is. It is not a deployable setting.

Metrics. The ictal class is the positive class. We report accuracy, balanced accuracy, precision, recall, F1, ROC-AUC and, for the classical models, average precision (PR-AUC). Results come from a single split and a single training run per model.

Environment. Code is Python with pandas, NumPy, SciPy, scikit-learn and TensorFlow.

IV. Results

A. Feature-Level Differences

Over all 1,673 segments, heart rate is clearly higher in ictal segments (mean 92.3 against 75.3 bpm), and MeanNN falls from 847.8 to 703.6 ms accordingly. This is the ictal tachycardia that often accompanies a seizure. The variability measures overlap heavily: SDNN and RMSSD have large standard deviations in both classes, and pNN50 is almost the same (27.5 against 28.9 %). Frequency-domain and entropy features are available for few ictal segments (295 and 273 against 793 and 780), so their means come from different subsets and should not be compared directly. These comparisons are descriptive only; we ran no significance tests.

B. Subject-Independent Classification

Table II gives the results on the 24 held-out subjects. The random forest has the best score on every metric except precision and specificity, where k-NN is higher (0.915 and 0.945) at the cost of a recall of 0.593. Fig. 4 shows the counts behind these numbers. The forest makes 9 false alarms and 10 misses among 182 segments, and its Matthews correlation coefficient is 0.791. Naive Bayes misses 50 of the 91 ictal segments. Fig. 3(a,b) show the ROC and precision–recall curves. The forest dominates the other three models over most of the range.

C. Within-Patient versus Subject-Independent

Repeating the experiment with the within-patient split, every model gains: the F1 of the random forest rises from 0.895 to 0.941 and its ROC-AUC from 0.939 to 0.982, while the F1 of the SVM, k-NN and naive Bayes rises to 0.892, 0.838 and 0.606. The Matthews coefficient of the forest goes from 0.791 to 0.881, and its errors drop from 19 of 182 segments to 22 of 371. The protocols also differ in test size (182 against 371 segments) and in which segments are tested, so the gap is a rough indication and not a controlled estimate. We have not yet repeated both protocols over many random splits.

D. Spread across Patients

The pooled scores hide differences between patients. On the 24 test patients, the random forest classified every segment correctly for 16 patients. The lowest per-patient accuracy was 0.50, three patients were below 0.75, and the mean over patients was 0.92 (median 1.0). The test patients contribute between 2 and 30 segments each, so a single wrong segment moves the accuracy of a small patient a lot. The numbers show that errors concentrate in a few patients, but they are too sparse to say why.

E. Feature Importance

Among the impurity importances of the random forest, the top feature is the mean successive Lorenz difference (0.140), followed by the mean QRS-width difference (0.068) and mean heart rate (0.044). Median heart rate and MeanNN also appear in the top ten, as do several beat-to-beat difference measures of QRS amplitude and R amplitude. The importance of heart rate agrees with Section IV-A. Impurity importance favours features with many distinct values and can be shared among correlated features, so we read the ranking as a rough guide and not as proof that a feature carries seizure information.

TABLE II
SUBJECT-INDEPENDENT TEST RESULTS (ICTAL CLASS POSITIVE; 24 HELD-OUT SUBJECTS).

Model	Accuracy (%)	Precision (%)	Recall (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	Balanced Accuracy (%)	MCC
Random Forest	89.56	90.00	89.01	89.01	90.11	89.50	89.56	0.791
SVM	84.62	85.39	83.52	83.52	85.71	84.44	84.62	0.692
KNN	76.92	91.53	59.34	59.34	94.51	72.00	76.92	0.575
Gaussian Naive Bayes	65.93	77.36	45.05	45.05	86.81	56.94	65.93	0.351

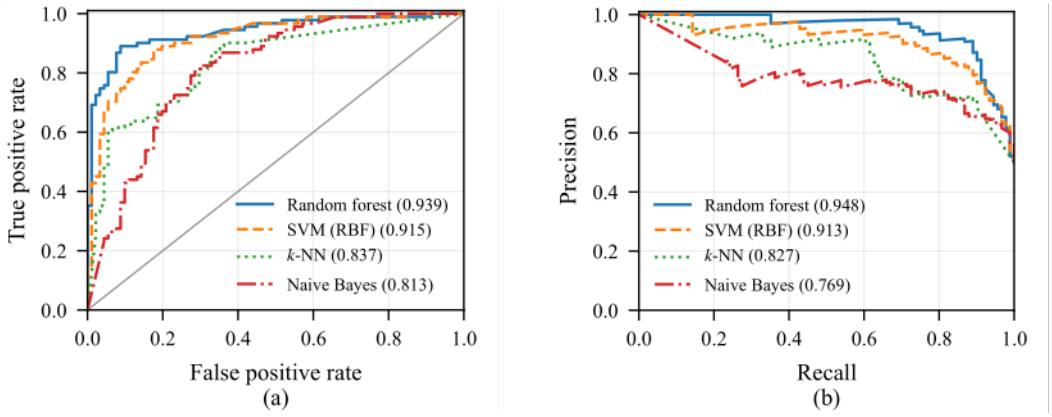


Fig. 3. (a) ROC and (b) precision–recall curves of the four classical models on the 24 held-out subjects. Area values are in parentheses.

F. Raw-Signal CNN

On the 6,315 test windows, 16.8 % of which are ictal, the baseline CNN reached an accuracy of 0.843 and a balanced accuracy of 0.565, but its ictal recall was only 0.145 (Fig. 5), and the accuracy is close to what always predicting the interictal class would give. Class weights and a validation-chosen threshold raised recall to 0.644 but cut precision from 0.653 to 0.263, which gives an F1 score of 0.373 (baseline 0.237) and a balanced accuracy of 0.639. The CNN is scored on windows with a different class balance, so these scores cannot be compared directly with the segment-level Table II. Even so, the ROC-AUC gap (0.719 and 0.697 against 0.939) is large.

During training of the improved network, the training AUC rose to 0.919 after 15 epochs, while the validation AUC peaked at 0.743 and ended at 0.695. The best validation loss (0.486) came at epoch 9. The baseline network stopped after 6 epochs. The widening gap between training and validation curves means the network fits its training patients and does not transfer to new ones.

V. DISCUSSION

A. What the Results Show

The feature-based random forest separated the two states of unseen patients far better than the raw-signal CNNs, and the within-patient split gave higher scores for every classifier. Heart rate is the one feature with a clear physiological reading here: it is higher in ictal segments, and it ranks among the top features of the forest. Most of the other leading features are beat-to-beat difference measures from the Lorenz plot and from QRS and R amplitudes. We cannot say from our experiments whether they reflect autonomic change during the seizure, movement artefact, or the length of the segment.

B. Relation to Earlier Work

Our scores are not comparable with published ones. The studies in Section II use other datasets, other segment definitions and, in several cases, a pre-ictal target. The one comparison we can draw is about the evaluation scheme. Kalousios *et al.* found a mean ROC-AUC of 0.823 with non-causal evaluation but 0.569 with pseudo-prospective evaluation [3]. We saw the same direction on a different task: scores fall when the test condition gets closer to real use. In our case the step is from a within-patient to a subject-independent split, and the random-forest ROC-AUC falls from 0.982 to 0.939. The drop is smaller than theirs, but our task is also easier, because the classes are ictal against distant interictal segments and not pre-ictal against interictal ones.

C. Why the CNN Fell Behind

We did not run ablations, so the following are hypotheses. First, the window normalisation removes absolute amplitude, which is one

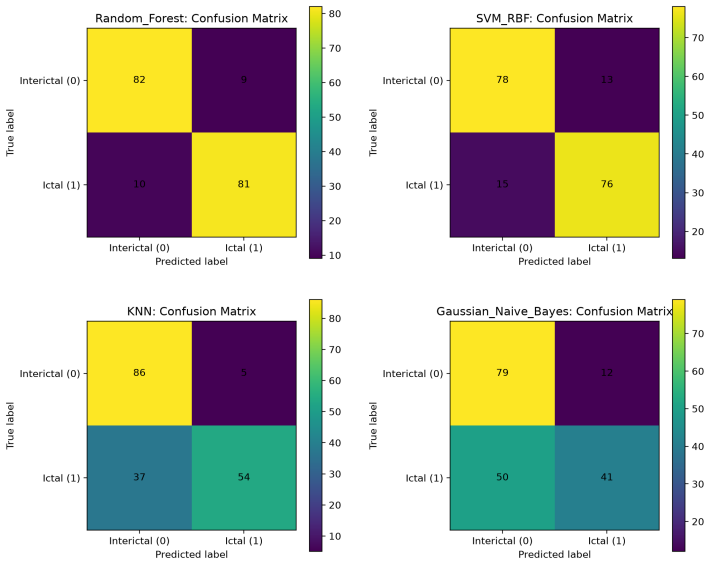


Fig. 4. Confusion matrices of the four classical models on the 182 test segments of the 24 held-out subjects (subject-independent split).

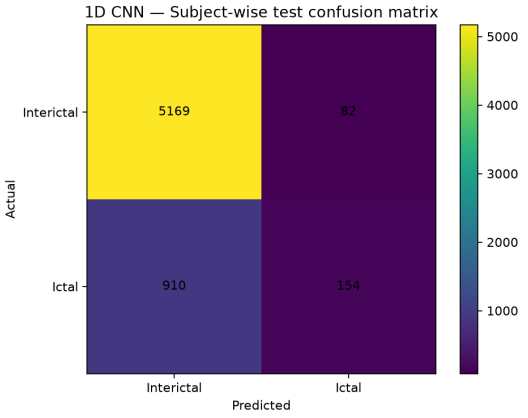


Fig. 5. Confusion matrix of the baseline 1D CNN on the 6,315 test windows of the 24 held-out subjects.

of the morphology cues the forest can use. Second, a 10 s window holds only a handful of beats, so a network on a single window cannot see the longer-range rhythm that features computed over a whole segment capture. Third, 119 patients may simply be too few for a network to learn a patient-independent representation from raw ECG, which fits the training–validation gap in Section IV-F. We also did not tune the CNN, and the two networks are a small sample of possible designs.

D. Limitations

Duration confound. Interictal segments are all 300 s long while ictal segments average 58.6 s, and 63.8 % of them are shorter than 60 s. We excluded length and count features, but many HRV statistics depend on the number of beats. The missing-value pattern is also informative: the filter removed the frequency-domain and nonlinear HRV features because they are missing for most ictal segments, and the median imputation we used for the remaining gaps could still let a classifier tell short from long. The random forest may therefore partly separate short from long segments. We have not yet rerun the experiment with interictal windows cut to the ictal length distribution, and until then the headline scores cannot be read as a measure of seizure-specific information.

Not prediction. The task separates segments that are known to be ictal from those at least 30 min away from a seizure. It does not test whether a pre-ictal state can be recognised in time, and no warning horizon or false-alarm rate is measured. Many studies in Section II address that harder problem, so their results do not compare with ours.

Evidence. Results come from one split and one run, without confidence intervals, and the test set has only 182 segments from 24 patients. The interictal label relies on the dataset annotations alone. The feature-based and CNN comparison also differs in unit of analysis and input length. We tuned no hyperparameters, and we made no comparison with published methods on identical data, because no such numbers exist for this segment definition. Finally, there is no trivial baseline such as a heart-rate threshold. Given Section IV-A, a simple heart-rate rule might score well, and a reviewer would reasonably ask for it.

E. Reproducibility and Data

SeizeIT2 is an open dataset. Our extraction, feature and model scripts, the per-segment predictions and the split membership lists are saved with the project, which allows the tables and figures above to be regenerated. Hardware and library versions are not reported in this paper.

VI. Conclusion

On single-lead SeizeIT2 ECG, a random forest on 82 features separated ictal from interictal segments of 24 unseen patients with an F1 score of 0.895 and a ROC-AUC of 0.939, while raw-signal CNNs reached an F1 score of at most 0.373 under the same split. A within-patient split gave higher scores for every classifier, so patient-mixed splits overstate what a deployed system would achieve. Heart rate was higher in ictal segments and was among the most important features.

Three things limit these conclusions: the duration difference between the classes, the single split and the absence of any prediction horizon. Our next steps follow from them. We will rerun the comparison with length-matched interictal windows and with repeated splits and confidence intervals, and add a heart-rate-only baseline. After that we plan to move to pre-ictal windows with a 15–30 min horizon and to report the seizure prediction horizon and the false-positive rate per hour. This planned work has not been done, and nothing in this paper should be read as a prediction result.

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